

## Supplementary Exercises

### Question : 1-8

Evaluate Determinant of given matrix by

(a) Cofactor Expansion

(b) and using Row operation; to intr 0 in Matrix.

These questions are same like

q<sub>n</sub> 23 and q<sub>n</sub> 28 solved in exercise 2.3

See Question : 23 & 28 in exercise 2.3

### Question : 9-

Evaluate determinant in question 3-6 using arrow technique

↳ This technique is valid only for  $2 \times 2$  &  $3 \times 3$  Matrix.

See example : 7 & Ques : 9-14 of exercise 2.1-

### Question : 11-12-

Use determinant, in to decide whether matrices in Exercise 1-4 & 5-8 are invertible.



Find out determinant if determinant comes out to be 0 then not invertible otherwise invertible.

### Question: 13-15

Find given determinant by any method:

Use any method that is easy or smart according to questions.

### Question: 16-

Solve for  $x$ :-

$$\begin{vmatrix} x & -1 \\ 3 & 1-x \end{vmatrix} = \begin{vmatrix} 1 & 0 & -3 \\ 2 & x & -6 \\ 1 & 3 & x-5 \end{vmatrix}$$

LHS:-

$$\begin{vmatrix} x & -1 \\ 3 & 1-x \end{vmatrix}$$

$$= x(1-x) + 3$$

$$= x - x^2 + 3$$



R.H.S:-

|   |     |       |
|---|-----|-------|
| 1 | 0   | -3    |
| 2 | $x$ | -6    |
| 1 | 3   | $x-5$ |

$$= 1 \begin{vmatrix} x & -6 \\ 3 & x-5 \end{vmatrix} + 0 + (-3) \begin{vmatrix} 2 & x \\ 1 & 3 \end{vmatrix}$$

$$= x(x-5) + 18 + 0 - 3(6-x)$$

$$= x^2 - 5x + 18 - 18 + 3x$$

$$= x^2 - 2x$$

Now:-

$$x - x^2 + 3 = x^2 - 2x$$

$$0 = x^2 + x^2 - 2x - x - 3$$

$$0 = 2x^2 - 3x - 3$$

$$\Rightarrow 2x^2 - 3x - 3 = 0$$

$$a=2, b=-3, c=-3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{3 \pm \sqrt{9 - 4(2)(-3)}}{4}$$

$$x = \frac{3 \pm \sqrt{33}}{4}$$

$$\boxed{x = \frac{3 + \sqrt{33}}{4}} \quad , \quad \boxed{x = \frac{3 - \sqrt{33}}{4}}$$



**Question: 17-24**

Use adjoint method to find inverse of matrices, if exist.

See Question: 19-23 of exercise 2.3

**Question: 34 (b)**

Use Result in (a) to find area of  $\Delta$  with vertices  $(3,3)$ ,  $(4,0)$ ,  $(-2,-1)$

$$\text{area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 3 & 3 & 1 \\ 4 & 0 & 1 \\ -2 & -1 & 1 \end{vmatrix}$$

$$\text{Area} = \frac{1}{2} \left[ 3 \begin{vmatrix} 0 & 1 \\ -1 & 1 \end{vmatrix} - 3 \begin{vmatrix} 4 & 1 \\ -2 & 1 \end{vmatrix} + 1 \begin{vmatrix} 4 & 0 \\ -2 & -1 \end{vmatrix} \right]$$

$$\text{Area} = \frac{1}{2} \left[ 3(0+1) - 3(4+2) + 1(-4-0) \right]$$

$$\text{Area} = \frac{1}{2} (3 - 18 - 4)$$



$$\text{Area} = \frac{1}{2} (-19)$$

$$\text{Area} = -9.5 \text{ m}^2$$

Since Area can't be negative so we ignore negative sign

$$\text{Area} = 9.5 \text{ m}^2$$

### Question : 35

use fact that

21375, 38798, 34162, 40223, 79154

are all divisible by 19 to show that

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 1 | 3 | 7 | 5 |
| 3 | 8 | 7 | 9 | 8 |
| 3 | 4 | 1 | 6 | 2 |
| 4 | 0 | 2 | 2 | 3 |
| 7 | 9 | 1 | 5 | 4 |

is divisible by 19 without directly evaluating determinant:

|   |   |   |   |       |
|---|---|---|---|-------|
| 2 | 1 | 3 | 7 | 21375 |
| 3 | 8 | 7 | 9 | 38798 |
| 3 | 4 | 1 | 6 | 34162 |
| 4 | 0 | 2 | 2 | 40223 |
| 7 | 9 | 1 | 5 | 79154 |

$$C_5 > 10^4 C_1 + 10^3 C_2 + 10^2 C_3 + 10^1 C_4 + C_5$$



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For Row  $R_1$ 

$$\begin{aligned} &= (10000 \times 2) + (1000 \times 1) + (100 \times 3) + (10 \times 7) + 5 \\ &= 20000 + 1000 + 300 + 70 + 5 \\ &= 21375 \end{aligned}$$

For Row  $R_2$ 

$$\begin{aligned} &= (10000 \times 3) + (1000 \times 8) + (100 \times 7) + (10 \times 9) + 8 \\ &= 30000 + 8000 + 700 + 90 + 8 \\ &= 38798 \end{aligned}$$

For Row  $R_3$ 

$$\begin{aligned} &= (10000 \times 3) + (1000 \times 4) + (100 \times 1) + (10 \times 6) + 2 \\ &= 30000 + 4000 + 100 + 60 + 2 \\ &= 34162 \end{aligned}$$

For Row  $R_4$ 

$$\begin{aligned} &= (10000 \times 4) + (1000 \times 0) + (100 \times 2) + (10 \times 2) + 3 \\ &= 40000 + 0 + 200 + 20 + 3 \\ &= 40223 \end{aligned}$$

For Row  $R_5$ 

$$\begin{aligned} &= (10000 \times 7) + (1000 \times 9) + (100 \times 1) + (10 \times 5) + 4 \\ &= 70000 + 9000 + 100 + 50 + 4 \\ &= 79154 \end{aligned}$$



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$$= \begin{vmatrix} 2 & 1 & 3 & 7 & 1125(19) \\ 3 & 8 & 7 & 9 & 2042(19) \\ 3 & 4 & 1 & 6 & 1798(19) \\ 4 & 0 & 2 & 2 & 2117(19) \\ 7 & 9 & 1 & 5 & 4116(19) \end{vmatrix}$$

$$= 19 \begin{vmatrix} 2 & 1 & 3 & 7 & 1125 \\ 3 & 8 & 7 & 9 & 2042 \\ 3 & 4 & 1 & 6 & 1798 \\ 4 & 0 & 2 & 2 & 2117 \\ 7 & 9 & 1 & 5 & 4166 \end{vmatrix}$$

Hence determinant is divisible by 19.

### Question: 36-

without Directly evaluating determinant  
Show that

$$\begin{vmatrix} \sin \alpha & \cos \alpha & \sin(\alpha + \beta) \\ \sin \beta & \cos \beta & \sin(\alpha + \beta) \\ \sin \gamma & \cos \gamma & \sin(\alpha + \beta) \end{vmatrix} = 0$$

$$= \begin{vmatrix} \sin \alpha \cos \alpha : \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ \sin \beta \cos \beta : \sin \beta \cos \alpha + \cos \beta \sin \alpha \\ \sin \gamma \cos \gamma : \sin \gamma \cos \alpha + \cos \gamma \sin \alpha \end{vmatrix}$$



$$= \begin{vmatrix} \sin \alpha & \cos \alpha & \cos \alpha \sin \beta \\ \sin \beta & \cos \beta & \cos \beta \sin \delta \\ \sin \delta & \cos \delta & \cos \delta \sin \delta \end{vmatrix} \quad c_2 - \cos \beta c_1$$

$$= \sin \delta \begin{vmatrix} \sin \alpha & \cos \alpha & \cos \alpha \\ \sin \beta & \cos \beta & \cos \beta \\ \sin \delta & \cos \delta & \cos \delta \end{vmatrix}$$

since  $c_2$  &  $c_3$  are proportional (equal)  
so determinant will be equal to 0

$$= \sin \delta (0)$$

$$= 0$$

$$\text{so LHS} = \text{RHS}$$